

Multiple Choice (10 marks)

1. What is the value of this python expression: `1 + 3 % 3`?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4
2. What is the output of `"abc"[::-1]` in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
3. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row
4. What is the output of the statement `"a" + "ab"` in Python 3?
 - (a) Error
 - (b) aab
 - (c) ab
 - (d) a ab
5. What is the output of `4*1**4` in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `x >> 1`, where `x` equals 8, evaluates to 3.
7. True or False: The `isupper()` function in Python 3 checks if all characters in a string are uppercase.
8. True or False: The `random()` function in Python 3 sets the integer starting value used in generating random numbers.
9. True or False: The condition `a < b` guarantees that the expression evaluates to true regardless of the values of `a` and `c`.
10. True or False: If a large Java program passes all tests, all its preconditions are guaranteed to be correct.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the RGB color model represents colors using red, green, and blue values. How many bits are required to accurately represent a color in this model, and why?
12. Explain how the exponential operator in Python 3 functions for calculating powers of numbers. Discuss its syntax, use cases, and any differences from other arithmetic operations.

End of Paper